

MD BILAYET HOSSAIN

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Mohishkundi-7052, Kushtia, Khulna, Bangladesh

EDUCATIONAL QUALIFICATIONS

Daffodil International University (DIU)

Bachelor of Science in Computer Science and Engineering

- CGPA: 3.77 /4.00

January 2022 -Present

Dhaka-1216, Bangladesh

Peoples College , Khash Mathurapur

Higher Secondary Certificate (HSC)

- GPA: 5.00 /5.00 Field: Science

May 2018 – June 2020

Kushtia, Khulna, Bangladesh

Pakuria Secondary School

Secondary School Certificate (SSC)

- GPA: 5.00 / 5.00 Field: Science

January 2012 -February 2017

Kushtia, Khulna, Bangladesh

INTERNATIONAL ACADEMIC EXPERIENCE

Academic Exchange Study Program (Japanese University Studies in Science & Technology (JUSST) Program).

The University of Electro-Communications, Tokyo, Japan

Duration: October 2024 – September 2025 (One Year)

- Successfully completed a one-year academic exchange program in Japan.
- Conducted research at the Yanai Laboratory under the supervision of Keiji Yanai.
- Participated in academic coursework, laboratory meetings, and collaborative research activities.
- Completed Japanese language training and intercultural academic programs.

PUBLICATIONS / RESEARCH PAPERS

I. Conference & Workshop Papers

1. **Hossain, M.B.**, Yuan, H., Akhy, S.A., Yanai, K. (2026). “*BengaliDiff: Diffusion Model for Few-Shot Bengali Font Generation.*” In: Proceedings on Document Analysis and Recognition – ICDAR 2025 Workshops. ICDAR 2025, Wuhan, China. Lecture Notes in Computer Science, vol 16226. Springer, Cham.
DOI: https://doi.org/10.1007/978-3-032-09371-4_7
2. Akhy, S.A., Shouno, H., Mojumdar, M.U., Hussain, T., **Hossain, M.B.**, Siddiquee, S.M.T. and Chakraborty, N.R. (2025), MPICam: A deep learning-based real-time medicinal plant identification system for rural healthcare in Bangladesh. In: Proceedings of Distributed Computing, HPC, Cloud Computing, Quantum Technologies, Embedded Systems, Modeling, and Simulation (CSCE 2025), Las Vegas, USA, 21–24 July.
DOI : https://doi.org/10.1007/978-3-032-22193-3_41
3. Anis, S.M., Mubin, M.M.R., Khandoker, M.A., **Hossain, M.B.**, Bhuiyan, M.N., Nur, F.N., Rashel, M.R., and Noori, S.R.H. (2026). “*Difficulty-Aware Cost-Constrained Routing of Vision-Language Models for Efficient Visual Question Answering.*” Submitted to the 29th International Conference on Computer and Information Technology (ICIT 2026). **Status: Submitted; Paper ID: 1995.**

I I. Journal Papers

1. Bhowmik, P., Rafi, M.A.H., Talukder, M.A., Shrabon, M.S.I., **Hossain, M.B.**, and Siam, M.M.A. (2026). “Multi-Paradigm XAI with Lightweight Explainable Convolutional Network and Feature Representation for Waste Classification.” Submitted to the *Turkish Journal of Electrical Engineering & Computer Sciences*.

Status: Submitted; Manuscript ID: TJEECS-2026-00985.

RESEARCH / PROFESSIONAL EXPERIENCE

Research Assistant (RA)

April 2026- Present

Health Informatics Research Lab (HIRL),
Daffodil International University, Bangladesh

Responsibilities:

-Contributing to AI and Health Informatics research by conducting literature review, developing and evaluating ML/DL models on biomedical datasets, and supporting paper writing, result analysis, and publication submissions.

Individual Research Project: Yanai Laboratory, The University of Electro-Communications, Tokyo, Japan

Title: BengaliDiff: Diffusion Model for Few-Shot Bengali Font Generation

Supervisor: Prof. Keiji Yanai

Duration: October 2024 – September 2025 (One Year)

- Conducted an individual research project under the JUSST exchange program
- Participated in laboratory meetings, discussions, and collaborative research activities
- **Presented my research work at the 19th International Conference on Document Analysis and Recognition 2025, Wuhan, China.**

SOFTWARE AND TECHNICAL KNOWLEDGE

- **Programming:** Python, C, C++, JavaScript, Dart, Bash
- **Web & Software Development:** HTML, CSS, React.js, TypeScript
- **Machine Learning & AI:** PyTorch, TensorFlow, Keras, Scikit-learn
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Seaborn, MATLAB
- **Tools & Platforms:** Git, TensorBoard, VS Code, CodeBlocks, Android Studio, Kaggle, Google Colab.
- **Research & Writing:** LaTeX (Overleaf), Zotero, Mendeley
- **Operating Systems:** Linux, Windows

UNDERGRADUATE THESIS (B.Sc ENGINEERING RESEARCH)

Title: Uncertainty - Guided Diffusion Network for Multi- Domain Biomedical Image Segmentation.

Description: Developing an uncertainty-guided diffusion framework for robust multi-domain biomedical image segmentation by integrating attention mechanisms, transformers, and explainable AI techniques.

Supervisor: Dr. Md. Fokhray Hossain

Professor, Dept. Of CSE, Daffodil International University (DIU), Bangladesh.

CERTIFICATIONS & ACHIEVEMENTS

A. International & Academic Programs

- International Exchange Study program “The Japanese University Studies in Science & Technology (JUSST)”, The University of Electro-Communications, Tokyo, Japan | 2024 – 2025
- Certificate of Commendation for Research Publications, Yanai Laboratory, UEC, Japan | 2025
- Certificate of Japanese Language Course “Elementary Japanese 1A” UEC, Japan | 2025
- Certificate of Global Healthcare Leadership Program, Daegu Health College, South Korea | 2022

B. MOOC, Workshop, and Competitions

- Completed Bootcamp on “Computer Vision & Deep Learning for Medical Data Analysis”, Daffodil International University (DIU), (HIRL Lab) | 2025

- Completed “AI Engineering Masterclass” Course offered by Department of CSE, DIU (April – August 2026)
- 1st Runner-up of “ModalX-AI Challenge – AI-Powered Multi-Modal Assessment System”, DIU | Jan 2026
- 1st Runner-up of “Research Poster Presentation” at 5th National Data Science Summit - DIU 2026
- Finalist Team on “AI Innovation Hackathon - DIU, 2026”

SELECTED RESEARCH & UNDERGRADUATE PROJECTS

1. ModalX-AI: Offline Video-Based Sentiment and Presentation Analysis System (March 2026 – Present)

Technologies: Python, Artificial Intelligence, Computer Vision, Video Analysis

- Developed an offline AI system for analyzing recorded presentation videos and generating sentiment and presentation-performance assessments while preserving user privacy through local processing.
- * Achieved **1st Runner-up** in the “**ModalX-AI Challenge – AI-Powered Multi-Modal Presentation Assessment System**”, Daffodil International University, January 2026. (GitHub: [ModalX-AI](#))

2. XAI-Driven Clinical Decision Support System for Interpretable Anemia Classification (April 2026- Present)

Technologies: Machine Learning, Random Forest, SHAP, Explainable AI, React, TypeScript

- Developed an explainable AI-based clinical decision support system for anemia classification using hematological parameters, achieving **98.98% accuracy** with Random Forest and integrating SHAP-based interpretation. (GitHub: [Anemia Classification](#)) & [Live Demo](#)
- * Achieved **1st Runner-Up in the Research Poster Presentation at the 5th National Data Science Summit, DIU 2026.**

3. BengaliDiff: Diffusion Model for Few-Shot Bengali Font Generation (October, 2024 – Present)

Technologies: Python, PyTorch, Diffusion Models, Computer Vision

- Developed a diffusion-based few-shot Bengali font generation framework with content–style representation learning and adversarial supervision.
- Published in ICDAR 2025 Workshops (Springer LNCS). [Paper Link](#) (GitHub: [BengaliDiff](#))

4. Dhaka Tesla Pool – Ride-Sharing System (September 2026 – Present)

Technologies: React, TypeScript, Vite, Node.js, Express.js, PostgreSQL, Docker, Vitest

- Developing a full-stack ride-sharing platform for Dhaka with secure session-based authentication, role-based access control, passenger ride requests, fare estimation, ride history and cancellation, PostgreSQL migrations, seeded demo data, and Dockerized deployment infrastructure. (GitHub: [Dhaka Tesla Pool](#))

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Executive Member, DIU Computer Programming Club (CPC) (February, 2023–2024)

- Responsibilities: Contributed to research discussions, paper reviews, and knowledge sharing.

Executive Member, Daffodil AI Club (December, 2025– September 2026)

- Responsibilities: Participated in AI research, literature review, and collaborative learning and Team management

REFERENCES

Dr. Md. Fokhray Hossain

Designation: Professor, Dept. Of CSE,

Daffodil International University (DIU)

Relationship: Undergrad Supervisor

Contact Information: Email:

drfokhray@daffodilvarsity.edu.bd

Phone: +8801713493250

Dr. Md. Zahid Hasan

Designation: Associate Professor, Dept. Of CSE,

Daffodil International University (DIU)

Relationship: Undergrad Co-Supervisor

Contact Information: Email: zahid.cse@diu.edu.bd

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